Financial Returns in Residential Real Estate Investments

Evidence from São Paulo

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Residential real estate is arguably the single largest component of household wealth in Brazil. Yet the tools available to analyze housing investments are surprisingly limited when compared to those the financial industry offers for equities, fixed income, or funds. Decisions involving hundreds of thousands—or millions—of reais are frequently made on the basis of broker opinions, neighborhood perceptions, and informal comparisons with nearby properties. It is what we might call *sentiment-based analysis*.

The contrast is striking: for a R$ 10,000 equity portfolio, investors have access to indices, historical returns, risk metrics, and benchmark comparisons. For a R$ 1 million apartment, there is often nothing of the sort.

This article seeks to contribute to changing that landscape. We present a Case-Shiller-type house price index for the residential market in São Paulo, built from administrative transaction records (ITBI). We compare this index to FipeZAP and to macroeconomic benchmarks—IPCA, CDI, and Ibovespa—for the period from 2010 to 2025.

This is, of course, a small effort in the face of a much larger task. Modernizing the way we think about and analyze real estate transactions—with the same rigor we already apply to other asset classes—is a long-term project that will require new data, new methodologies, and above all, a cultural shift. But every project starts with a first step.

Why is it so hard to measure housing returns?

The total return on a real estate investment can be decomposed into four components:

$$\textbf{Total Return} = \textbf{Price Appreciation} + \textbf{Rental Income} - \textbf{Maintenance Costs} - \textbf{Taxes}$$

Price appreciation is the difference between the sale price and the purchase price. **Rental income** represents the cash flow generated by the property over the holding period (or, for owner-occupiers, the rent they avoid paying). **Maintenance costs** include condominium fees, renovations, and upkeep. And **taxes** encompass property tax (IPTU), transfer tax (ITBI), and capital gains tax upon sale.

The problem is that reliable data on all of these components practically do not exist in an integrated form in Brazil. Rents available on platforms are *asking* prices, not contracted ones. Maintenance and renovation costs are nearly impossible to observe at scale. And the appreciation component—which is the most tractable—faces its own challenges: properties are unique, transact infrequently, and exhibit enormous spatial heterogeneity (even within the same building).

Moreover, the most widely used data in Brazil—such as the FipeZAP index—are based on listings, not actual transactions. This can generate a distorted picture of the market, especially during periods of low liquidity, when sellers maintain high asking prices even as the market adjusts.

In this article, we focus exclusively on the **price appreciation** component. The index we construct does not capture rental income, nor costs or taxes. It is one piece—an important one—of a larger puzzle.

A Case-Shiller index for São Paulo

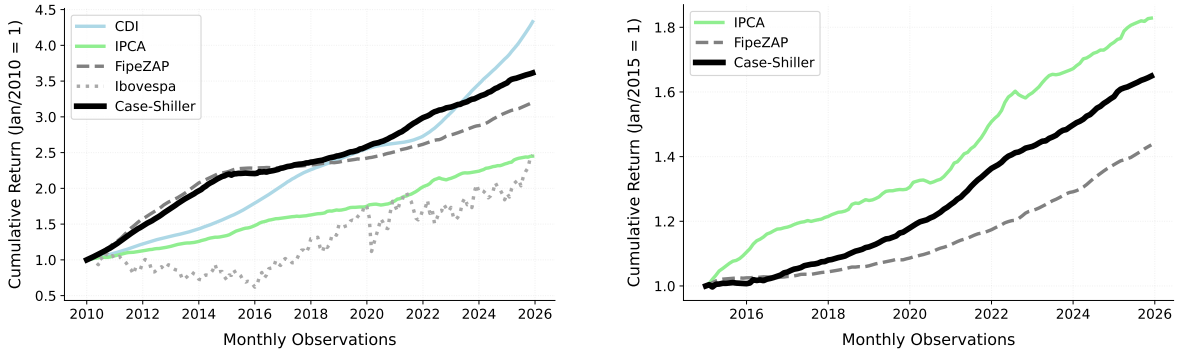
The index we present follows the classic methodology of Case and Shiller (1987, 1989), widely used in the United States and in other countries. The central idea is simple: instead of comparing average property prices across different periods (which confounds price changes with changes in the composition of properties sold), we track the *same* property over time, comparing its price at the second sale with its price at the first. Each repeat-sale pair gives us one individual return observation. The aggregate index is constructed by regression: we

estimate a set of monthly coefficients that jointly best explain the thousands of individual returns observed. The result is a time series that reflects the general price trend, controlling for the fact that different properties transact at different times.

The data come from ITBI (*Imposto sobre Transmissão de Bens Imóveis*—a real estate transfer tax) records published by the São Paulo municipal government. Because it is a transaction tax, the recorded values reflect prices actually paid—not listing prices. The database is extensive: approximately 2.6 million transactions involving roughly 1.5 million unique properties, covering the period from 2006 to 2025.

From this universe, we identify properties that appear in at least two transactions (the so-called *repeat sales*). Pairs with holding periods shorter than two years are excluded to reduce noise from atypical transactions (such as speculative flips). After this filtering, approximately 309,000 repeat-sale pairs remain, corresponding to roughly 263,000 distinct properties—a robust basis for estimation. The estimation uses generalized least squares (GLS), with a heteroskedasticity correction proportional to the holding period, and a three-month moving average smoothing. The final index is normalized to 1 in January 2010.

Cumulative returns: 2010–2025



(a) Since January 2010. **(b)** Since January 2015.

Figure 1: Cumulative returns for different asset classes. Series normalized to 1 at the base date. *Source: ITBI/São Paulo City Hall, FipeZAP, IBGE, BCB, B3. Authors' calculations.*

Figure 1 summarizes the central results. In the left panel, from January 2010 to December 2025, CDI accumulated the highest nominal return (4.3x), followed by Case-Shiller (3.6x), FipeZAP (3.2x), Ibovespa (2.5x), and IPCA (2.5x). Deflating by IPCA, the Case-Shiller index implies a cumulative real gain of approximately 48%, while FipeZAP implies roughly 31%.

The right panel isolates the 2015–2025 period—and reveals a different story. Over this window, IPCA accumulated approximately 83%, while Case-Shiller rose 65% and FipeZAP only 44%. Both housing indices fell *below* inflation during this subperiod, suggesting negative real capital gains.

Case-Shiller vs. FipeZAP: same trend, different conclusions

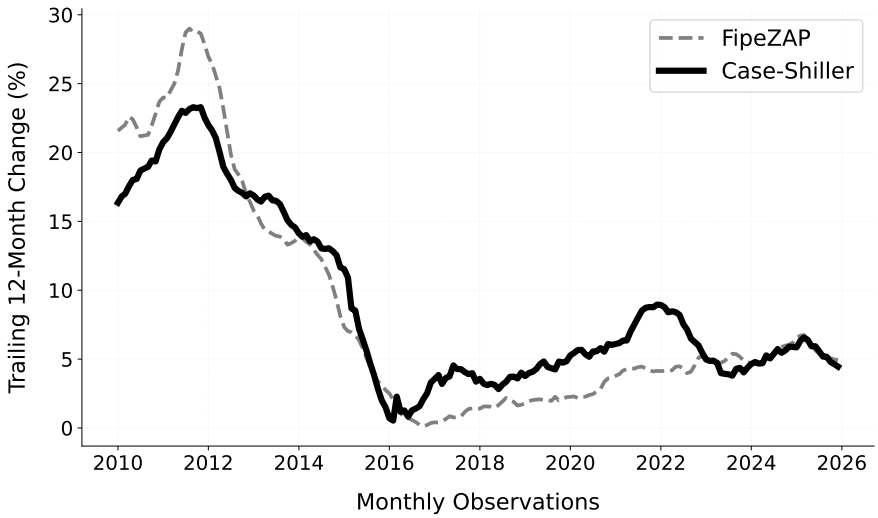


Figure 2: Trailing 12-month price appreciation: Case-Shiller vs. FipeZAP. *Source: ITBI/São Paulo City Hall, FipeZAP. Authors' calculations.*

Figure 2 shows the year-over-year appreciation rates for both indices. Until 2013, they move together, with appreciation rates above 15% per year. From 2014 onward, the indices diverge: FipeZAP decelerates more rapidly, with appreciation near zero between 2016 and 2019. Case-Shiller also decelerates, but maintains higher rates throughout the period.

This divergence is not surprising. FipeZAP measures *asking* prices, which tend to be sticky on the downside: sellers are reluctant to reduce listing prices even when the actual market is adjusting. Case-Shiller, based on effective transactions, may capture price dynamics more faithfully. This does not mean one index is "right" and the other "wrong"—they measure different things.

The cumulative difference, however, is significant: 21 percentage points between 2015 and 2025. Depending on which index one uses, the conclusion about housing returns changes materially.

Did housing lose to inflation?

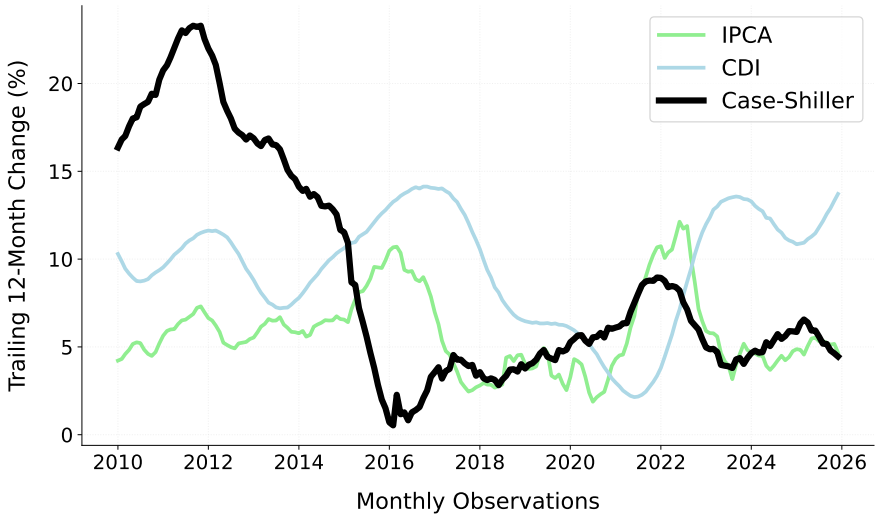


Figure 3: Housing price appreciation vs. IPCA and CDI (trailing 12-month change). *Source: ITBI/São Paulo City Hall, IBGE, BCB. Authors' calculations.*

Recent press reports have claimed that housing in São Paulo has not appreciated in real terms over the past decade. The claim is not entirely without foundation—but it is fragile, for at least three reasons.

First, it depends on the base date. Starting from 2015, both indices do indeed fall below inflation. But starting from 2010, the result reverses comfortably.

Second, it depends on the index. Case-Shiller implies a much smaller real shortfall than FipeZAP.

Third, and most importantly: capital gains are only one component of total return. Rental income—captured by neither index—may fully or partially offset the real loss in appreciation. To claim that housing was a "bad investment" based solely on capital gains is, at best, premature.

Figure 3 also reveals an important pattern: a historically negative correlation between interest rates (CDI) and housing appreciation. During high-rate periods (2015–2016, 2022–2023), appreciation decelerates; during monetary easing (2017–2020), prices tend to accelerate. This relationship is not surprising—lower rates reduce mortgage costs, increase demand, and lower the opportunity cost of holding capital in real estate—but it is relevant for thinking about the future. If the monetary cycle ahead is indeed more accommodative, with gradual reductions

in the policy rate, the historical pattern suggests that the housing market may benefit.

The heterogeneity hidden by aggregate indices

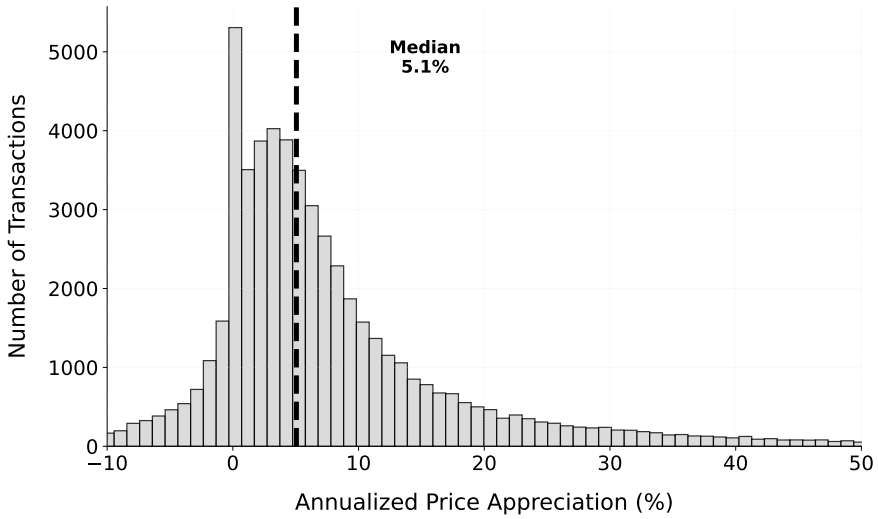


Figure 4: Distribution of annualized price appreciation per property (repeat sales, 2015–2025). The dashed line indicates the median of 5.1%. *Source: ITBI/São Paulo City Hall. Authors' calculations.*

Perhaps the most important result of this study is in Figure 4. It shows the distribution of annualized appreciation rates for individual properties, based on repeat sales with purchases between 2015 and 2019—precisely the worst period for the São Paulo housing market, when aggregate indices appear stagnant in real terms.

The distribution is wide and right-skewed. The median is 5.1% per year, but a substantial share of transactions show returns of 10%, 20%, or more.

The analogy with the equity market is instructive. Over the 2010–2025 period, the Ibovespa accumulated a nominal return of approximately 146%—roughly matching inflation. Would it be reasonable to conclude that investing in stocks was a bad decision? Obviously not: the aggregate index conceals enormous dispersion across individual stocks. Those who chose well—whether through fundamental analysis, timing, or active management—obtained returns far above the index. The same reasoning applies to real estate: just as there is a wide cross-section of returns across stocks, there is also a wide cross-section of returns across properties. The investor who selects well—by location, by property type, by entry timing—can achieve results far from the average.

An important caveat: the individual returns in Figure 4 reflect only transaction price appreciation. They do not account for renovation, maintenance, or improvement expenditures between purchase and sale. To the extent that part of the appreciation reflects investment in the property, the net gain would be lower.

Conclusion and outlook

Three central messages emerge from the analysis.

The choice of index matters. Case-Shiller and FipeZAP agree on broad trends, but diverge enough to change conclusions about real returns. Users of housing data should understand what each index measures—and its limitations.

The claim that housing did not appreciate is fragile. It depends on the base date, on the index used, and it ignores relevant components of total return—especially rental income.

Aggregate indices hide enormous heterogeneity. Even during periods of apparent stagnation, many individual properties generated substantial returns.

For investors, the implications are clear. The residential real estate market offers real opportunities for value creation—but capturing them increasingly requires a quantitative and disciplined approach. Careful asset selection, grounded in data and return metrics, can make the difference between a mediocre investment and an exceptional one. The index presented here is a step in that direction—but much remains to be done: integrating rental data, maintenance costs, geographic disaggregation, and building risk metrics. The analytical infrastructure for real estate needs to catch up with the sophistication that already exists for other asset classes—and that is exactly the path we are pursuing.

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